3. option (b)

$k = A\, e^{-\left(\frac{E_a}{RT}\right)}$

$\ln k = \ln A - \left(\frac{E_a}{R}\right)\left(\frac{1}{T}\right)$

this equation is in the form of a straight line equatoion

$y = c + m\, x$

a plot of $\ln k$ vs $\left(\frac{1}{T}\right)$ is a straight line with negative slope

4. option(d)

For a first order reaction

$t_{1/2} = \frac{0.693}{k}$

$t_{1/2}$ does not depend on the initial concentration and it remains constant (whatever may be the initial concentration)

$t_{1/2} = 2.5$ hrs

5. option(c)

$$Rate = \left(\frac{-1}{2}\right)\frac{d[NH_3]}{dt} = \frac{d[N_2]}{dt} = \left(\frac{1}{3}\right)\frac{d[H_2]}{dt}$$

$$\left(\frac{1}{2}\right)k_1[NH_3] = k_2[NH_3] = \left(\frac{1}{3}\right)k_3[NH_3]$$

$$\left(\frac{3}{2}\right)k_1 = 3k_2 = k_3$$

$1.5\, k_1 = 3k_2 = k_3$

6. option(c)

Given :

At low pressure the reaction follows first order, therefore

$Rate\ \alpha [\text{reactant}]^1$

$Rate\ \alpha\ (\text{surface area})$

At high pressure due to the complete coverage of surface area, the reaction follows zero order.

$Rate\ \alpha [\text{reactant}]^0$

Therefore the rate is independent of surface area.

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